Hult International Business School

(MFIN - Data Extraction & Visualization – Fall 2024)

Professor Thomas Kurnicki

Extracting Data Using SQL and Loading into Data Viz

Portfolio Analysis (Customer# 128, Bojana Popovic)

Renan Peres

Part #1 – Data Extraction (MySQL View)

```sql
USE invest;
CREATE VIEW RenanPeres AS (

/*
SQL VIEW: This VIEW below contains the daily pricing (adjusted) details for the securities that customer #128 (Bojana Popovic) posses in his portfolio (as
            The data contains information for the periods BETWEEN '2019-09-08' AND '2022-09-09'.
            The VIEW also contains the filters and dimensions that will be useful to answerd the questions for the PART #1 (Q. 1-4) of the Assignment as we
*/

SELECT
    pd.date,
    cd.customer_id,
    cd.full_name,
    ad.account_id,
    ad.main_account AS main_account_id,
    hc.ticker,
    sm.security_name,
    sm.sec_type,
    sm.major_asset_class,
    sm.minor_asset_class,
    hc.quantity,
    pd.value AS adj_closing_price
FROM invest.customer_details cd
JOIN invest.account_dim ad ON ad.client_id = cd.customer_id
JOIN invest.holdings_current hc ON hc.account_id = ad.account_id
JOIN invest.security_masterlist sm ON hc.ticker = sm.ticker
JOIN invest.pricing_daily_new pd ON pd.ticker = sm.ticker
WHERE
    cd.customer_id = '128'
    AND pd.price_type = 'Adjusted'
    AND pd.date BETWEEN '2019-09-08' AND '2022-09-09'
);

SELECT *
FROM RenanPeres
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

date	customer_id	full_name	account_id	main_account_id	ticker	security_name	sec_type	major_asset_class	minor_asset_class	quantity	adj_closing_price
2019-09-09	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.055443
2019-09-10	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.127087
2019-09-11	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.008224
2019-09-12	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.085566
2019-09-13	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.114872
2019-09-16	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.652189
2019-09-17	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.355852
2019-09-18	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.285025
2019-09-19	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.301306
2019-09-20	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.244316
2019-09-23	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.369692
2019-09-24	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.247574
2019-09-25	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.175119
2019-09-26	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.097776
2019-09-27	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	18.024506
2019-09-30	128	Bojana Popovic	16001	160	BCI	abrdn Bloomberg All Commodity Strategy K-1 Fr...	etf	commodities	precious_metals	339	17.934954

<u>Question 1</u>: Using data for your client tickers from the VIEW, what is the most recent 12 months, 24 months, 36 months return for each of the securities (and for the entire portfolio)?

```sql
WITH price_history AS (
    SELECT DISTINCT
        rp.date,
        rp.ticker,
        rp.quantity,
        rp.adj_closing_price,
        NULLIF(LAG(rp.adj_closing_price, 250) OVER (
                PARTITION BY rp.ticker
                ORDER BY rp.date
                ), 0) AS prev_12m,
        NULLIF(LAG(rp.adj_closing_price, 500) OVER (
                PARTITION BY rp.ticker
                ORDER BY rp.date
                ), 0) AS prev_24m,
        NULLIF(LAG(rp.adj_closing_price, 750) OVER (
                PARTITION BY rp.ticker
                ORDER BY rp.date
                ), 0)  AS prev_36m
    FROM RenanPeres rp
    WHERE CAST(rp.adj_closing_price AS DECIMAL) != 0
),
ror AS (
    SELECT
        date,
        ticker,
        quantity,
        adj_closing_price,
        (adj_closing_price-prev_12m)/prev_12m as ror_12m,
        (adj_closing_price-prev_24m)/prev_24m as ror_24m,
        (adj_closing_price-prev_36m)/prev_36m as ror_36m
    FROM price_history
),

-- Part 1: Most Recent Return for each of the Securities
sec_ror AS (
    SELECT DISTINCT *
    FROM ror
    WHERE date = '2022-09-09'
)

-- Part 2: Return for the whole Portfolio
SELECT
    'Portfolio Return' as ticker,
    SUM(quantity * ror_12m) / NULLIF(SUM(CASE WHEN ror_12m IS NOT NULL THEN quantity END), 0) as ror_12m,
    SUM(quantity * ror_24m) / NULLIF(SUM(CASE WHEN ror_24m IS NOT NULL THEN quantity END), 0) as ror_24m,
    SUM(quantity * ror_36m) / NULLIF(SUM(CASE WHEN ror_36m IS NOT NULL THEN quantity END), 0) as ror_36m
FROM sec_ror;
```

date	ticker	quantity	adj_closing_price	ror_12m	ror_24m	ror_36m
2022-09-09	ARB	177	26.999001	0.024863384451867546	0.04655403519652693	0.07245291251054271
2022-09-09	ARB	369	26.999001	0.024863384451867546	0.04655403519652693	0.07245291251054271
2022-09-09	BCI	339	27.42	0.21169275554885253	0.6437305706190894	0.4995877774298914
2022-09-09	BIDU	418	138.429993	0.16405983289707074	-0.15208870979078498	-0.48110805338146806
2022-09-09	BIDU	758	138.429993	0.16405983289707074	-0.15208870979078498	-0.48110805338146806
2022-09-09	BIL	525	91.470001	0.0038565342735350124	0.0032312337083233392	0.011697353731785626
2022-09-09	BNDX	831	48.709999	-0.12126709210858323	-0.1202387640616859	-0.09821137414853538
2022-09-09	CDNS	939	174.679993	0.06104590715404512	0.6599827807664322	1.5767810948992314
2022-09-09	CEG	146	87.510002	NULL	NULL	NULL
2022-09-09	CTA	445	28.91	NULL	NULL	NULL
2022-09-09	DBA	112	20.67	0.11488667125746117	0.40135593220338994	0.357997779111905
2022-09-09	DBC	982	25.389999	0.30138385443362364	0.9560862095531586	0.6772904313384719
2022-09-09	DIS	446	115.18	-0.3773380771399735	-0.12237126172008292	-0.15299652848442594
2022-09-09	DJP	422	35.23	0.247079646017699	0.7774974772956608	0.5805293144670561
2022-09-09	FLSP	436	21.375	0.042329551688573885	0.029458552481580202	NULL
2022-09-09	GIGB	310	45.294998	-0.15883411413166554	-0.1446118076548664	-0.05569823337278298
2022-09-09	GLDM	331	34.07	-0.04512331838565021	-0.12416456945169309	0.14405637528353346
2022-09-09	GOOGL	411	110.650002	-0.22259498786025714	0.44158119658063993	0.7953190605605809
2022-09-09	IAUM	977	17.139999	-0.04432673790503148	NULL	NULL
2022-09-09	LBAY	824	27.583	0.2168925849537317	NULL	NULL
2022-09-09	MARB	213	20.190001	0.02539359952292537	0.04163452724730559	NULL
2022-09-09	MSVX	263	24.25	-0.07466481879456875	0.03167760776723483	NULL
2022-09-09	OTIS	828	73.400002	-0.17291193821242182	0.24845778897995524	NULL
2022-09-09	RINF	343	33.330002	0.14242439168132745	0.34226152649397773	0.35599517244591944
2022-09-09	RLY	440	28.709999	0.12506234181871753	0.4784969324663674	0.37075415926741095
2022-09-09	ROK	987	255.380005	-0.17710512458305297	0.1751467096184974	0.6482606806634921
2022-09-09	SGOL	415	16.450001	-0.044160258231275874	-0.12453432014186694	0.14402955699283684
2022-09-09	TSCO	847	200.919998	0.01056630271156793	0.4664590764343994	1.2373095512774548
2022-09-09	UNG	829	27.889999	0.5408839226519335	1.218774781225139	0.22539538664323364

ticker	ror_12m	ror_24m	ror_36m
Portfolio Return	0.04674713065146581	0.31065832717167874	0.4013171842362576

<u>Key Findings</u>: Bojana's portfolio has a **weighted average return** of **4.7%** (12M), **31.06%** (24M), and **40.13%** for his whole portfolio. The top 3 asses by the daily returns in the most recent 12-month window are CEG, UNG, and DBC. We are going to see how the returns correlate with risk (sigma) it in the next question.

<u>Question 2</u>: What is the most recent 12months sigma (risk) for each of the securities? What is the average daily return for each of the securities?

```sql
WITH prev_1d AS (
    SELECT
        date,
        ticker,
        adj_closing_price,
        NULLIF(LAG(adj_closing_price, 1) OVER (
                PARTITION BY ticker
                ORDER BY date
            ), 0) AS prev_1d
    FROM RenanPeres rp
    WHERE CAST(adj_closing_price AS DECIMAL) != 0
),

ror AS (
    SELECT *,
        (adj_closing_price-prev_1d)/prev_1d as ror_1d
    FROM prev_1d
),

-- Part 1: What is the daily average return for each of the Securities?
avg AS (
    SELECT
        ticker,
        AVG(ror_1d) as avg_daily_ror_12m
    FROM ror
    WHERE ror_1d IS NOT NULL AND date BETWEEN '2021-09-08' AND '2022-09-09'
    GROUP BY ticker
),

-- Part 2: What is the Most Recent 12months signma(risk)?
std AS (
    SELECT
        ticker,
        STDDEV(ror_1d) as std_12m
    FROM ror
    WHERE ror_1d IS NOT NULL AND date BETWEEN '2021-09-08' AND '2022-09-09'
    GROUP BY ticker
)

SELECT DISTINCT
    ror.ticker,
    avg.avg_daily_ror_12m,
    std.std_12m
FROM ror
JOIN avg ON avg.ticker = ror.ticker
JOIN std ON std.ticker = ror.ticker
ORDER BY
    avg.avg_daily_ror_12m DESC,
    std.std_12m ASC
```

ticker	avg_daily_ror_12m	std_12m
CEG	0.005142053261819878	0.03264409737273617
UNG	0.0033025821361914926	0.04683974245898776
DBC	0.001252104985197199	0.01620940344762782
DJP	0.00109924635071074	0.016186209303207882
CTA	0.00109309531675919	0.009256101599352083
BCI	0.0009416696850798482	0.014629578079599559
LBAY	0.0007760980109473401	0.009024750761550763
RINF	0.0006161728806647736	0.01222617646785709
CDNS	0.0005353798735355029	0.02489973966162182
RLY	0.0005290829313683462	0.010380190023224115
DBA	0.0003896102694721437	0.009742731944966315
TSCO	0.00034172201942492535	0.020805148772411783
FLSP	0.00022885700332168294	0.008577927444266966
MARB	0.00011202676543486992	0.0025628197964832154
ARB	0.00009344914083876046	0.0028118618210684716
BIDU	0.000051419410872349...	0.029937078969005455
BIL	0.00001559129851824126	0.00011812055332306...
SGOL	-0.0001409530983331358	0.008576598561576547
IAUM	-0.00014140027677453...	0.008552977986358408
GLDM	-0.00014477908865761...	0.008592447757061762
MSVX	-0.0003277025169136195	0.005053508819835222
BNDX	-0.0004943763499136051	0.0037192118846714633
ROK	-0.0005765025248660053	0.02100805067037716
OTIS	-0.0005876419308859439	0.01638779274060762
GIGB	-0.0006334942028670151	0.005445866623959341
GOOGL	-0.0008164567368235113	0.021369062459606927
DIS	-0.0016555024893089095	0.01967702929254265

<u>Key Findings</u>: As shows in the screenshot above, **CEG** was the **best performing asset** within the most recent 12-month period, giving a daily return of **0.5%/day** and a risk of **0.032** in the period. Perhaps CEG was the **best performing ticker** of all the assets included in our database. The **second-best performing** ticker (which Bojana does not process) is **OXY**. We will look at some of the Top 3 performing tickers that the customer does not possess in the next question.

<u>Question 3</u>: Suggest adding a new investment to your portfolio - what would it be and how much risk (sigma) would it add to your client?

```sql
-- Part 1: Extract Historical data for the Securities (NOT Included in the (
WITH price_history AS (
    SELECT
        pd.date,
        pd.ticker,
        sm.security_name,
        sm.sec_type,
        sm.major_asset_class,
        sm.minor_asset_class,
        pd.value AS adj_closing_price,
        NULLIF(LAG(pd.value, 1) OVER (
                PARTITION BY pd.ticker
                ORDER BY pd.date
                ), 0) AS prev_1d,
        NULLIF(LAG(pd.value, 250) OVER (
                PARTITION BY pd.ticker
                ORDER BY pd.date
                ), 0) AS prev_12m
    FROM pricing_daily_new pd
    JOIN security_masterlist sm ON pd.ticker = sm.ticker
    WHERE pd.price_type = 'Adjusted'
        AND pd.date BETWEEN '2021-09-08' AND '2022-09-09'
        AND pd.ticker NOT IN (Select DISTINCT ticker from RenanPeres)
),

-- Part 2: Returns
ror AS (
    SELECT
        date,
        ticker,
        security_name,
        sec_type,
        major_asset_class,
        minor_asset_class,
        adj_closing_price,
        (adj_closing_price-prev_1d)/prev_1d as ror_1d,
        (adj_closing_price-prev_12m)/prev_12m as ror_12m
    FROM price_history
),

-- Part 3: Sigma
sigma AS (
    SELECT
        ticker,
        AVG(ror_1d) as avg_daily_ror_12m,
        STDDEV(ror_1d) as std_12m
    FROM ror
    WHERE ror_1d IS NOT NULL
    GROUP BY ticker
),

-- Part 4: Ranking Securities by Return and Risk
ranked_securities AS (
    SELECT
        r.ticker,
        r.security_name,
        r.sec_type,
        r.major_asset_class,
        r.minor_asset_class,
        s.avg_daily_ror_12m,
        s.std_12m,
        ROW_NUMBER() OVER (
            PARTITION BY r.major_asset_class
            ORDER BY s.avg_daily_ror_12m DESC, s.std_12m ASC
        ) as rank_within_type
    FROM ror r
    JOIN sigma s ON r.ticker = s.ticker
    WHERE r.date = '2022-09-09' AND r.ror_12m > 0
)

-- Part 5: Selecting Best Performing Security in for each major_asset_class
SELECT *
FROM ranked_securities
WHERE rank_within_type = 1
ORDER BY
    sec_type,
    rank_within_type;
```

ticker	security_name	sec_type	major_asset_class	minor_asset_class	avg_daily_ror_1...	std_12m	rank_within_type
OXY	Occidental Petrol...	common_share	equity	large_cap	0.004	0.036	1
PFIX	Simplify Interest ...	etf	alternatives		0.002	0.026	1
UCO	ProShares Ultra ...	etf	commodities	precious_metals	0.003	0.049	1

<u>Key Findings</u>: As shown in the screenshot above, next to CEG, **OXY** was the second-best performing asset within the most recent 12-month period. Giving a daily return of **0.4%/day** and with a risk of **0.036** in the same period. For that reason, I would suggest the client to add OXY (which is an large cap) to his portfolio. By adding OXY to the portfolio, the average unlevered sigma (risk) would go from **0.014** to **0.015.**

<u>Question 4</u>: Calculate risk adjusted returns for each of your securities by following this formula: AVG(returns for ticker)/STD(returns for ticker). Which of the securities is best from the rest (with highest risk adjusted returns), why?

```sql
WITH price_history AS (
    SELECT DISTINCT
        rp.date,
        rp.ticker,
        rp.quantity,
        rp.adj_closing_price,
        NULLIF(LAG(rp.adj_closing_price, 1) OVER (
                PARTITION BY rp.ticker
                ORDER BY rp.date
                ), 0) AS prev_1d
    FROM RenanPeres rp
    WHERE CAST(rp.adj_closing_price AS DECIMAL) != 0
),
ror AS (
    SELECT
        date,
        ticker,
        quantity,
        adj_closing_price,
        (adj_closing_price-prev_1d)/prev_1d as ror_1d
    FROM price_history
),
stats AS (
    SELECT
        ticker,
        AVG(ror_1d) as avg_ror_1d,
        STDDEV(ror_1d) as std_ror_1d
    FROM ror
    WHERE ror_1d IS NOT NULL
    GROUP BY ticker
)
SELECT
    ticker,
    (avg_ror_1d / std_ror_1d) as adj_ror_1d
FROM stats
ORDER BY
    adj_ror_1d DESC
;
```

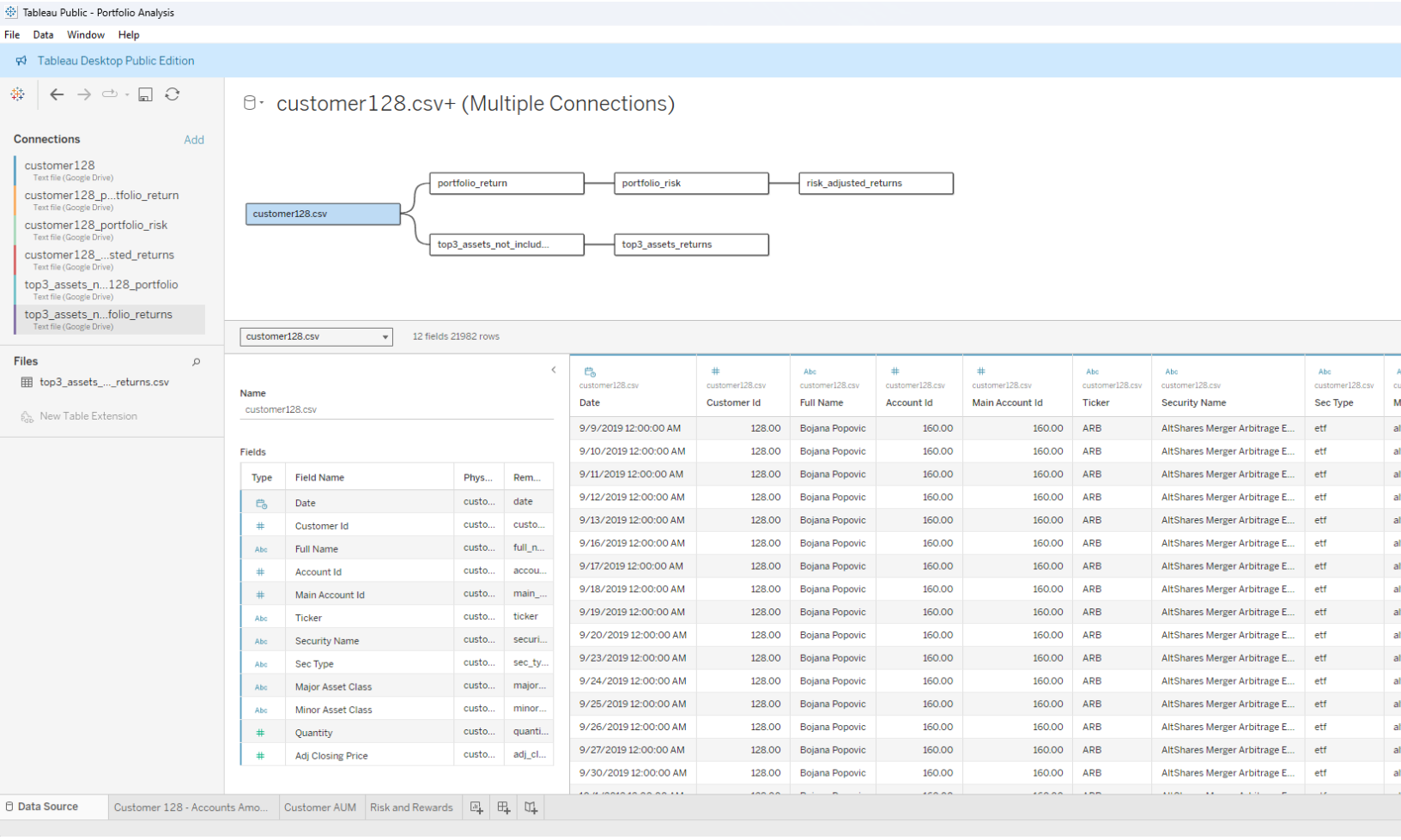
ticker	adj_ror_1d
CEG	0.1575186228342291
BIL	0.14612547293477857
CTA	0.11809456767800663
LBAY	0.09418581812839777
CDNS	0.06248692137983389
DBC	0.0570096664226634
TSCO	0.055683985460540625
DBA	0.054215130342413057
DJP	0.05349679060009036
OTIS	0.05279786188046116
BCI	0.051011078263571784
GOOGL	0.049654686309897396
RINF	0.04447858099955686
RLY	0.04282073984757232
ROK	0.04059221853550728
ARB	0.0375583457589873
UNG	0.026156821948324024
GLDM	0.0222578360855012
SGOL	0.022249509399657204
BIDU	0.019355084028212677
MARB	0.005016795556160164
MSVX	0.0010496546900063978
DIS	0.0008472846818484087
FLSP	-0.006447800563826762
IAUM	-0.009676505848555976
GIGB	-0.01013880094828059
BNDX	-0.04639087838139039

<u>Key Findings</u>: As mentioned in question 2, **CEG** was the **best performing ticker** overall in the client's portfolio. After adjusting the historical daily return by the risk, CEG alone gave an adjusted return of **15%**.

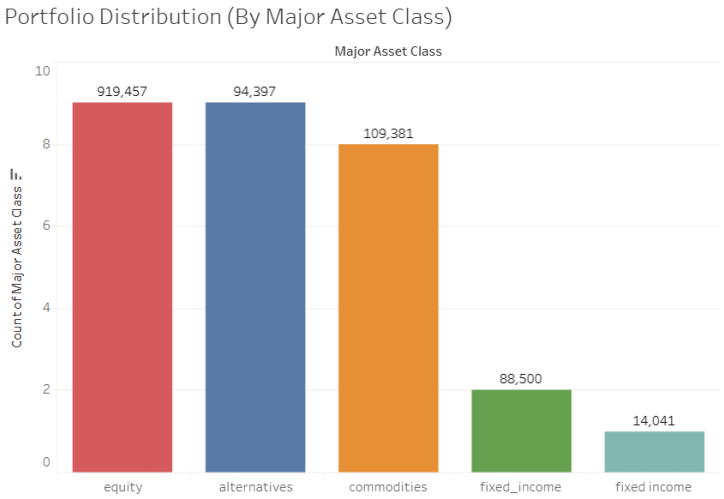
Part #2 – Data Visualization (Tableau)

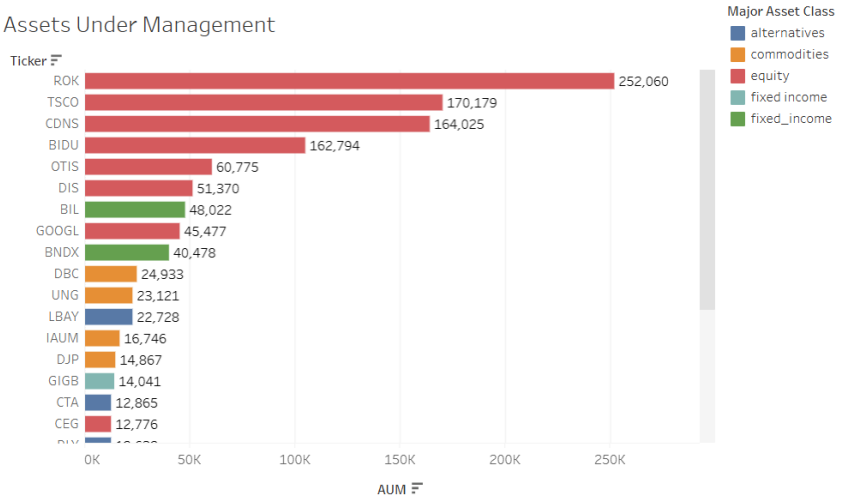
Datasets and Data Model

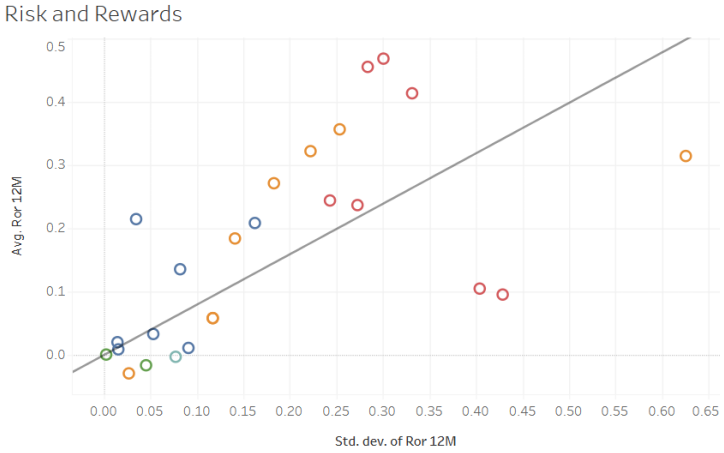
- Customer 128 (Bojana Popovic) Portfolio (Exported from SQL View)
- Portfolio Returns (Exported from Question 1)
- Portfolio Risks (Exported from Question 2)
- Best Performing Assets not included in Customer's Portfolio (Exported from Question 3)
- Portfolio Risk Adjusted Returns (Exported from Question 4)

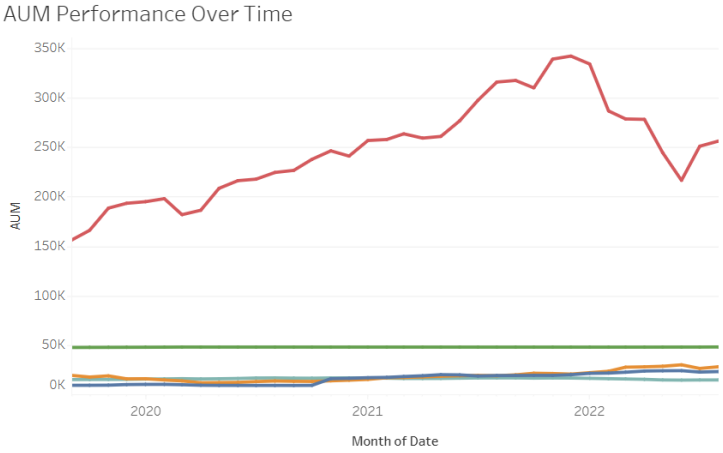


Dashboard


Portfolio Distribution (By Major Asset Class)
Major Asset Class
Count of Major Asset Class
10
8
6
4
2
0
919,457
94,397
109,381
88,500
14,041
equity
alternatives
commodities
fixed_income
fixed income

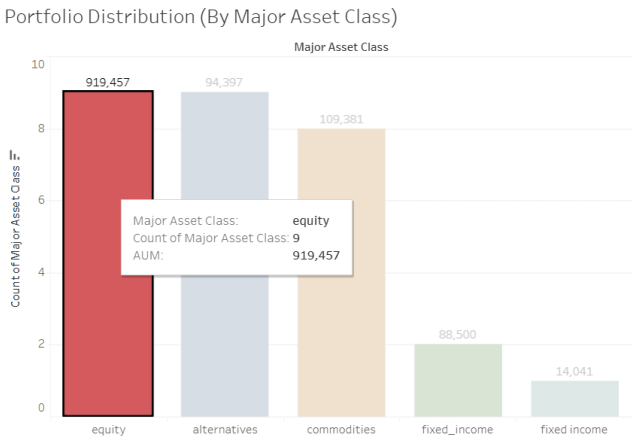

Assets Under Management
Major Asset Class
alternatives
commodities
equity
fixed income
fixed_income
Ticker
ROK 252,060
TSCO 170,179
CDNS 164,025
BIDU 162,794
OTIS 60,775
DIS 51,370
BIL 48,022
GOOGL 45,477
BNDX 40,478
DBC 24,933
UNG 23,121
LBAY 22,728
IAUM 16,746
DJP 14,867
GIGB 14,041
CTA 12,865
CEG 12,776
0K
50K
100K
150K
200K
250K
AUM

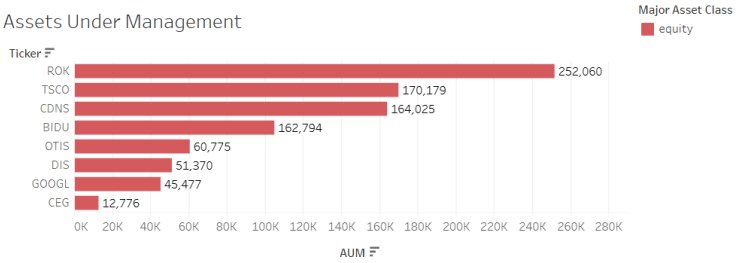

Risk and Rewards
Avg. Ror 12M
0.5
0.4
0.3
0.2
0.1
0.0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Std. dev. of Ror 12M

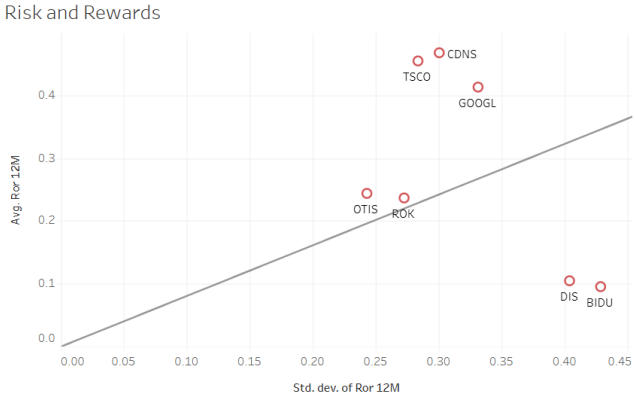

AUM Performance Over Time
AUM
350K
300K
250K
200K
150K
100K
50K
0K
2020
2021
2022
Month of Date

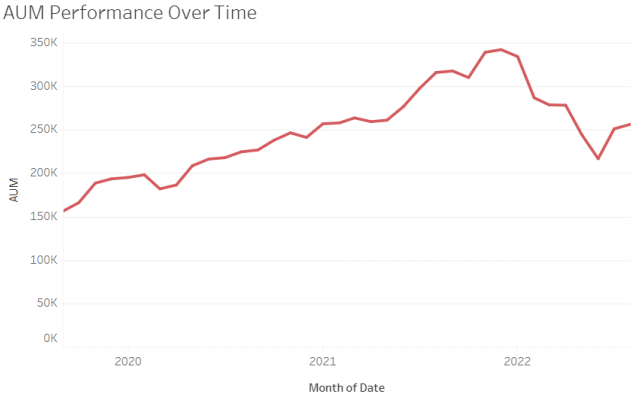
Key Findings and Business Insights:

1. **Portfolio Distribution**: As shown in the first distribution in the bar chart below (from right to left), the customer is mostly invested in equity and alternatives, possessing a count of 9 assets in each group. The distribution chart also shows the total amount of his portfolio as of the last day of the data range, September 09, 2021. Although the distribution for equity and alternatives is the same, 9 assets, we can clearly see that this money is mostly concentrated in equity ($919,457) against $94,397 in alternatives. The third largest distribution, which is commodities, has 8 assets under management and an invested amount of $109,301. Fixed income comes right after with 2 assets and a total of $88,500. Lastly, fixed income has a total of $14,031 in assets under management.

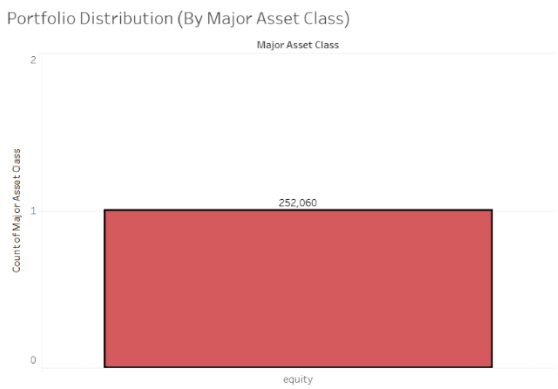


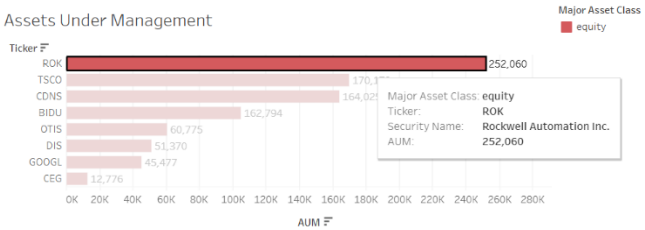


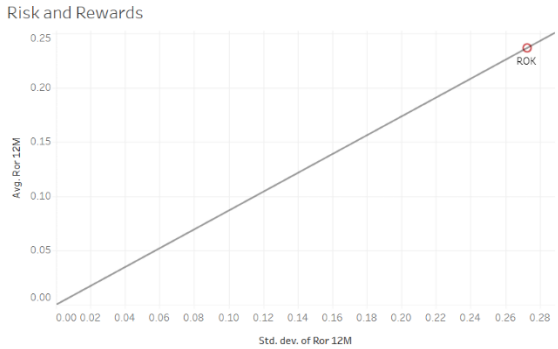


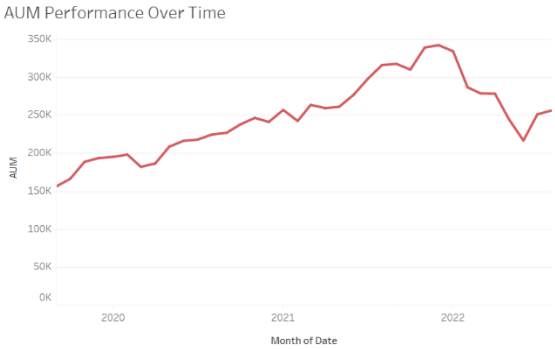


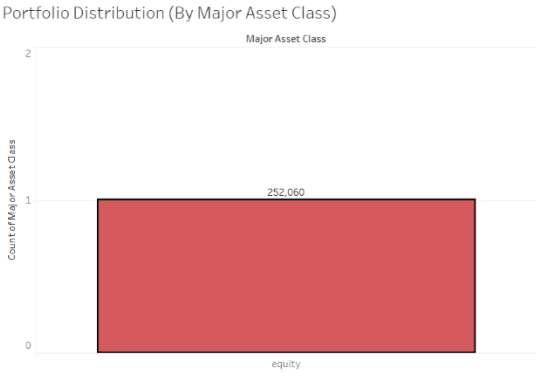
2. **Assets Under Management**: The customer's AUM is mostly represented by his allocation in ROK, being $252,060 of the whole portfolio as of August 2022. ROK has seen a steady increase in its price and peaked in December 2021, hitting a total of $341,963 in that period. Since then, ROK has had a drop in the AUM, but has seen a pullback again in the last year.

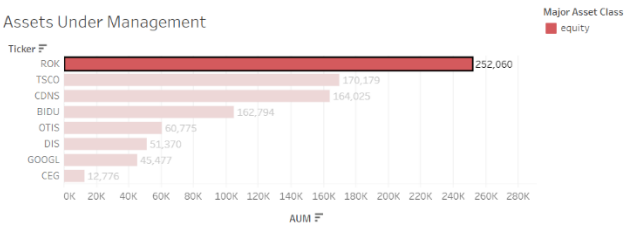


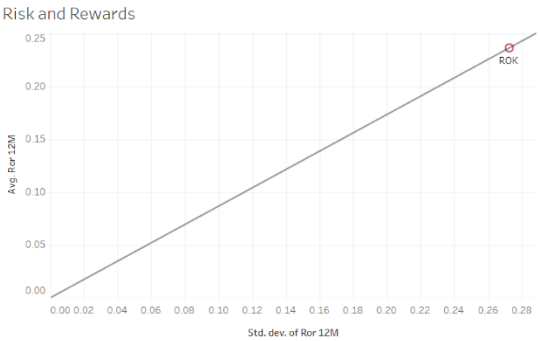


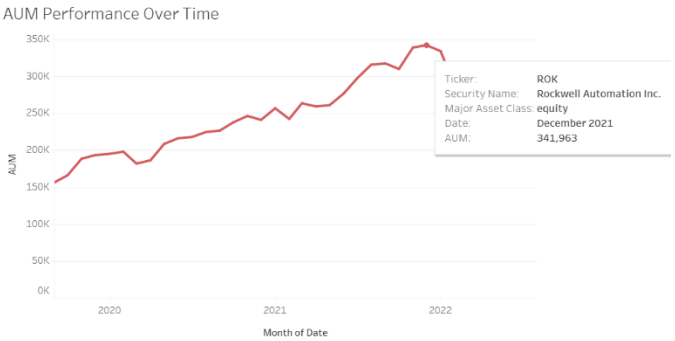
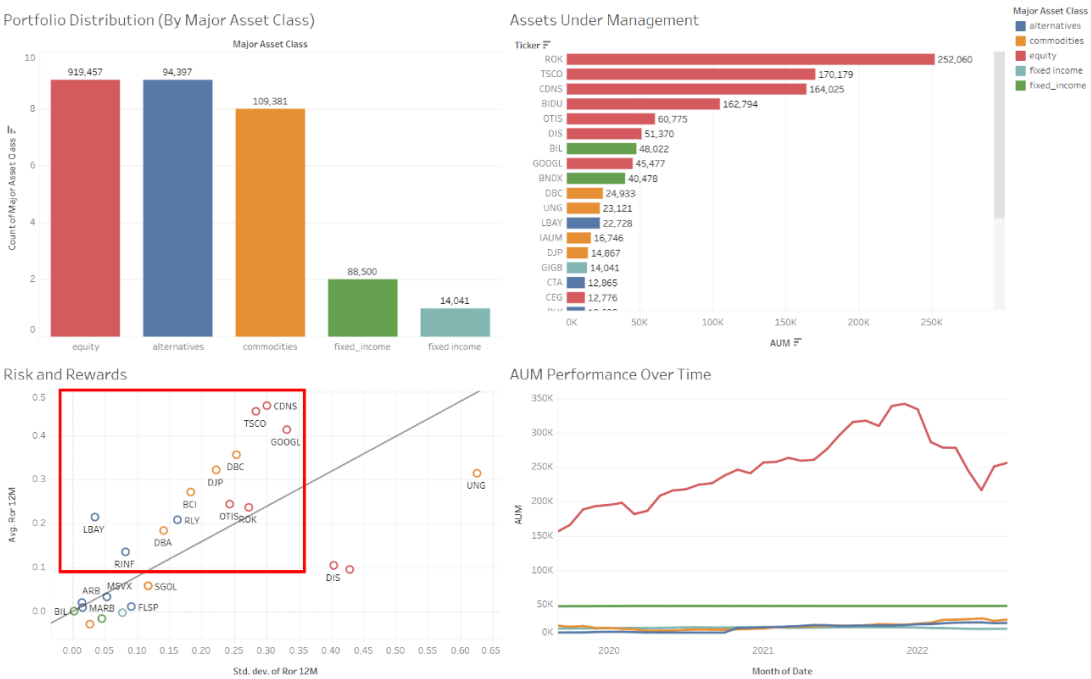
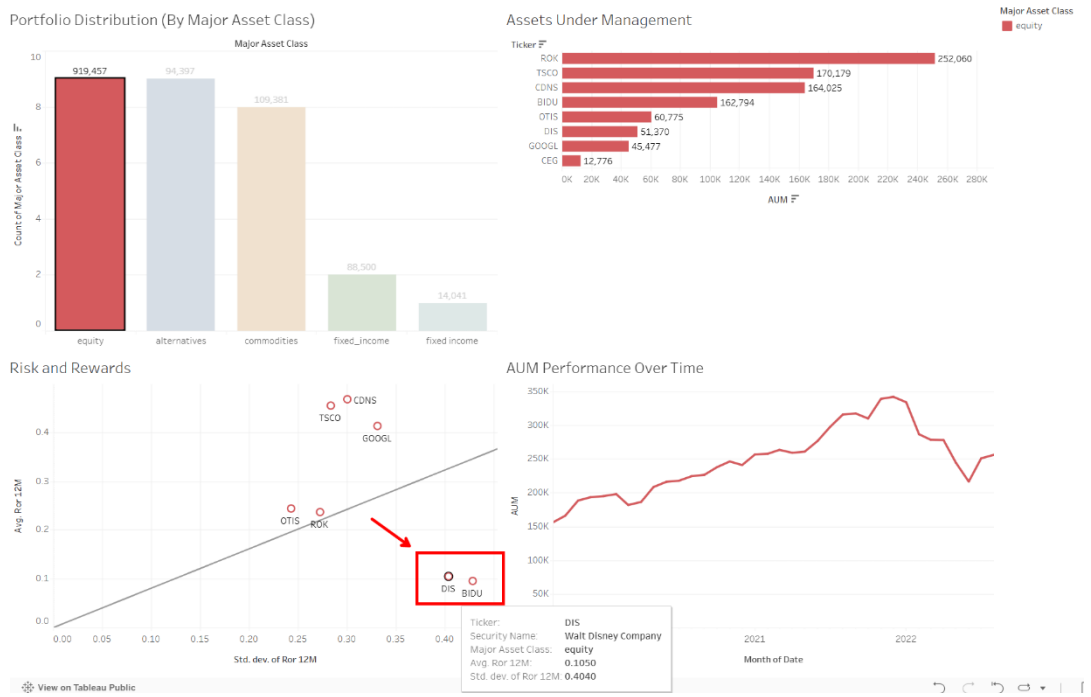












3. **Portfolio Risks and Rewards**: Overall, the portfolio has a good balance between the risks and returns. These can be explained by the 4th chart. Essentially, all the assets that are above the trendline and as far to the right as possible are the ones that give the clients the best returns and lower risks. On the other hand, the assets below the trendline present a return below the average and a risk higher than the average. For example, in the screenshot below, the underperforming assets for their risks are Walt Disney Company and Baidu. That means that the client should **divest their positions in these assets** and **concentrate on the upper left side of the chart**.